Midterm Project - TFY4240 Electromagnetic Theory

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1 Introduction

In this project we will solve the problem of finding the electric potential and fields of a long, square hollow tube with sides of length L where one side has a arbitrary potential, using numerical methods. The other three sides will have constant zero potential. There are no charges in the tube, so from Gauss' law we get the problem of solving Laplace equation in 2d with our boundary conditions.

2 Theory

2.1 Laplace equation & boundary conditions

The form and relevance of the Laplace equation

$$\Delta V(x, y) = 0 \tag{1}$$

is assumed to be somewhat familiar. In our context it arises from Gauss' law in its differential form, where V is the electric potential. We are under the boundary constraints of

$$\begin{cases} V(x = 0, y) = 0 \\ V(x = L, y) = 0 \\ V(x, y = 0) = 0 \\ V(x, y = L) = V_0(x), \end{cases} . \tag{2}$$

where V_0 is arbirtary. In order to implement our numerical methods with more ease, we will perform a change of variables into unitless variables. We set $\xi = x/L$ and $\zeta = y/L$. We futher consider the potential V_0 unitless and in the range $(-1, 1)$ by dividing by its maximum absolute value.

2.2 Laplace equation by seperation of variables

In order to solve (1) for certain boundary condidtions, we employ the method of seperation of variables. By setting $V = \Xi(\xi)Z(\zeta)$ in (1), we get $\Xi'' = -k^2\Xi$ and $Z'' = k^2 Z$, for some real k, where the sign of the right side has been chosen to fit our specific boundary conditions. Solving these we get the general solution

$$V = (Ae^{k\zeta} + Be^{-k\zeta})(C\cos(k\xi) + D\sin(k\xi).$$

This is done in more detail in example 3.3 in [1]. From $V(\xi = 0, \zeta) = 0$ we get $C = 0$, and from $V(\xi = 1, \zeta) = 0$ we get $k = n\pi$, for some natural n. From $V(\xi, \zeta = 0) = 0$ we get $A = -B$. The general solution satisfiying our three first boundary conditions is then

$$V = C_n \sinh(\pi n\zeta)\sin(\pi n\xi)$$

. We write the general solution as a sum of these solutions and get

$$V(\xi, \zeta) = \sum_{n=1}^{\infty} C_n \sinh(\pi n\zeta)\sin(\pi n\xi). \tag{3}$$

The objective is to find C_n's so that the fourth boundary condition holds. We notice that V is written as a Fourier-series in sines for a given ζ. In particular, for $\zeta = 1$, we get

$$V_0(\xi) = \sum_{n=1}^{\infty} C_n \sinh(\pi n) \sin(\pi n \xi).$$

By using that the Fourier-terms are mutually orthogonal under integration over the interval $(0, 1)$, we get by integrating both sides that

$$C_n = \frac{2}{\sinh(\pi n)} \int_0^1 V_0(\xi) \sin(\pi n \xi) d\xi. \tag{4}$$

Our objective in this project is to numerically approximate these integrals and to include a suitable number of coefficients as to approximate the boundary conditions satisfactorily using the series.

3 Results

The method of approximating V by numerically approximating the coefficients 4 and including a suitible number of terms from the seires 3, was done for the potentials

$$V_0 = \begin{cases} \sin(m\pi\xi) \\ 1 - (\xi - 0.5)^4 \\ \theta(\xi - 0.5)\theta(0.75 - \xi) \end{cases} . \tag{5}$$

The numerical integration was done with the scipy.integration function "quad", and 180 terms were included in 3. The approximated potentials at the boundary $\zeta = 1$ is shown in Figure 1, together with the actual boundary functions.

The sum of the difference between the boundary functions and the approximation, for different numbers of terms, is shown in Figure 2. The boundary points for the power-four potential was removed in calculating error since the fixed boundary points of the approximation differ from the potential at the actual boundary.

The electric field in the entire tube was calculated from the potential and plotted as a quiverplot in Figure 3.

4 Discussion/Conclusion

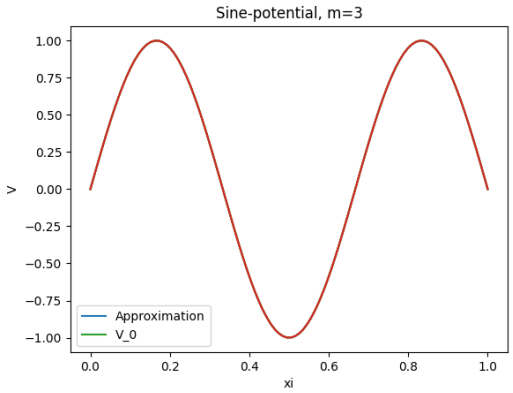
As seen in the evolutions of errors in Figure 2, the sine-potential converges exactly towards the actual solution. This is illustrated in Figure 1 (a) as well. This is to be expected, as the boundary is expressed exactly as a term in 3. As such, one would expect to see the error drop to 0 at the inclusion of 3 terms, which is what Figure 2(a) shows. However, the power four- and the Heaviside-potentials do not converge. The boundary conditions imposed on the power-four potential from the other sides of the tube causes a discontinuity at the corners. The approximation does not become exact, even when this endpoint is removed. This is due to the Gibbs phenomenon, where the jump in function value forces the truncated series to overshoot, and sometimes oscillate. This can be seen in Figure 1(b).

The Heaviside-potential does not converge either, for the same reason. The is clearly seen in Figure 1 (c),whre the approximation overshoots at both sides of the discontinuities.

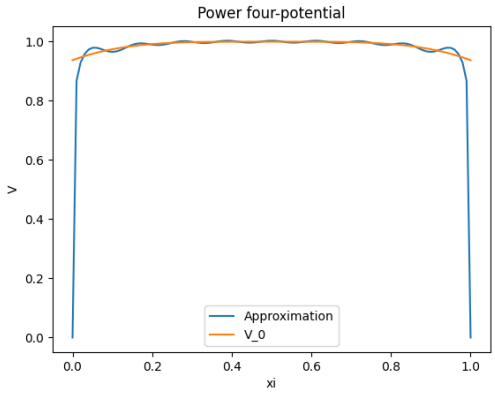
Figure 3 shows the electric fields calculated from the gradient of the approximated V, for the same three boundary potentials. We see a large electric field around the discontinuities in potential in (b) and (c), which is to be expected. We also see that the electric field necessarily is not zero at the other boundaries, but should have a direction perpendicular to them. The exception is at corners. The field subsiding away from $\zeta = 1$ is expected, as it becomes influenced by the other boundaries.

References

[1] D.J. Griffiths. *Introduction to Electromagnitism.* Prentice-Hall Inc., 1999.



(a)



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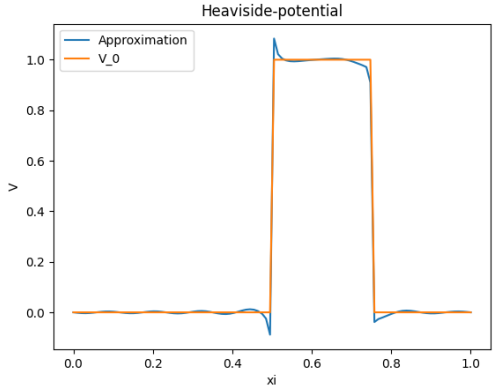
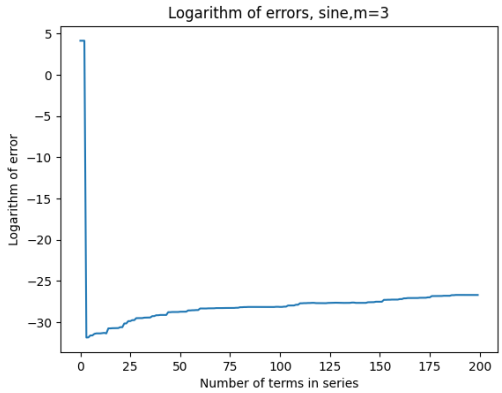


(c)

Figure 1: Boundary potentials and approximated boundary potential using (3) with 180 terms, for three different V_0.



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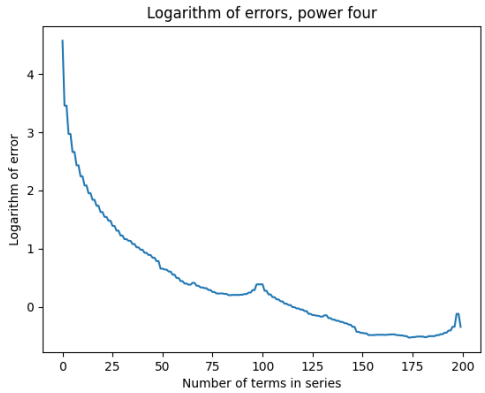

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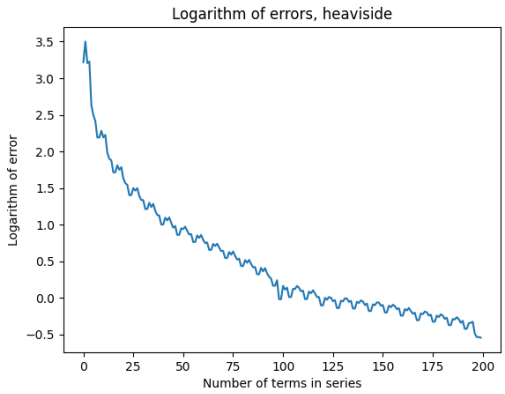
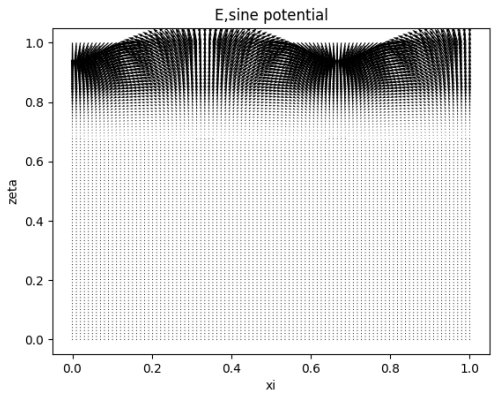
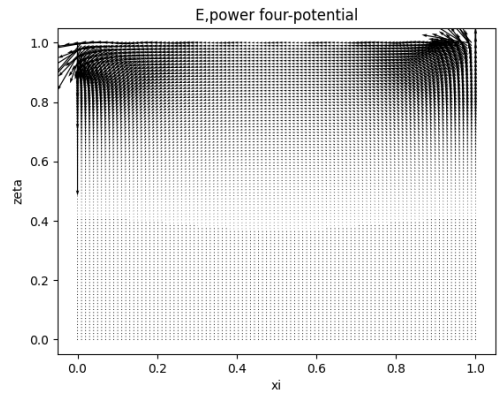


(c)

Figure 2: Sum of error in all points between approximated and acutal boundary potential for different number of terms included in (3).



(a)



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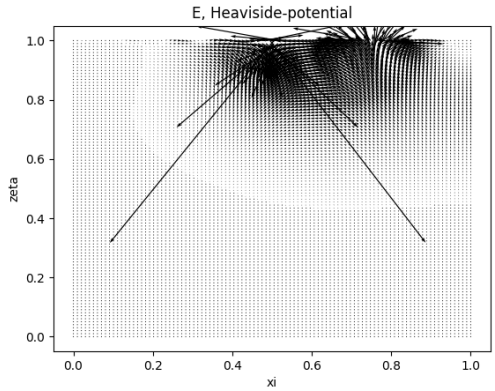

(c)

Figure 3: Quiverplot of electric field inside the tube, for three different boundary potentials at $\zeta = 1$.